Atelier: An Automated Analog Circuit Design Framework via Multiple Large Language Model-Based Agents

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***Abstract*—This article introduces Atelier, a large language model (LLM)-based framework for analog circuit design to address the issues of data scarcity and the substantial domain-specific knowledge required in this field. Atelier integrates general-purpose LLMs with a high-quality, compact knowledge base to fulfill the considerable knowledge requirements of analog circuit design, obviating the need for extensive domain-specific training or fine-tuning. The knowledge base is meticulously curated to be task-oriented and encapsulates critical information from pertinent literature within user-defined templates, leveraging the LLMs' capabilities in text comprehension and summarization. The framework comprises several LLM agents, structured in a graph-of-thoughts architecture, with each agent specialized in a distinct task in analog circuit design, including circuit analysis, topology selection, topology modification, parameter tuning, and design decision. This collaborative multiagent system, enriched with access to the compact knowledge base and advanced mechanisms, such as self-reflection, backtracking, and tool integration, automates the analog circuit design process. It significantly enhances design quality and efficiency while ensuring interpretability. Experimental results highlight Atelier's superiority over state-of-the-art black-box methods, general-purpose LLMs, and LLM-based methods, demonstrating notable improvements in success rates, design quality, and runtime.**

***Index Terms*—Analog circuit design, design automation, graph of thoughts (GoT), large language model (LLM).**

I. Introduction

ANALOG circuits play a crucial role in enabling the functionality of advanced technologies, such as portable electronics and the Internet of Things, by providing essential signal processing, data conversion, and communication interfaces that ensure seamless integration between the digital and physical worlds. The design of analog circuits is inherently intricate, necessitating a combination of domain-specific physical intuition, mathematical analysis, and experiential expertise. In the case of commonly utilized analog circuit modules, experts often need to invest substantial time and effort into iterative trial-and-error procedures to tailor these components for specific applications. As a result, there is a significant demand for automated analog circuit design to reduce the design cycle and maintain high design quality. This article focuses on the topology synthesis of analog circuits, which includes both topology design and parameter tuning (also known as *sizing*).

Analog circuit design automation has garnered considerable research attention. In the context of analog circuit sizing, black-box optimization methods, such as Bayesian optimization (BO) [1], [2], [3] and reinforcement learning (RL) [4], [5], [6], have demonstrated success due to their effectiveness in low-dimensional, continuous parameter spaces. However, the synthesis of analog circuit topology poses a more complex challenge, as it involves designing both the circuit topology and its parameters within a joint space of discrete topologies and continuous parameters. Existing methods for analog circuit topology synthesis can be categorized into two approaches: 1) topology selection and 2) topology generation. The former involves selecting topologies from a predefined library, either by developing heuristic rules [7], [8], [9] or by training data-driven machine learning (ML) models [10], [11]. However, these methods are constrained by a limited range of compensation structures and require significant effort to develop selection rules, train ML models, and construct the topology library. Due to these limitations, topology generation methods have emerged as the mainstream approach for topology synthesis. These methods typically utilize black-box optimization techniques, such as evolutionary algorithms [12], [13], [14], BO with graph embedding techniques [15], [16], [17], and RL [18], [19], [20] to explore optimal circuit topologies and parameters in the expansive design space. However, these black-box approaches face significant challenges, particularly due to the lack of domain-specific knowledge on analog circuit design.

Received 12 September 2024; revised 18 March 2025; accepted 19 May 2025. Date of publication 23 May 2025; date of current version 6 January 2026. This work was supported in part by the National Key R&D Program of China under Grant 2020YFA0711900 and Grant 2020YFA0711901, and in part by the National Natural Science Foundation of China (NSFC) under Project 92473206, Project 62474050, Project 62304052, Project 92473207, Project 62474051, Project 92373207, and Project 62090025. This article was recommended by Associate Editor S. Ozev. *(Corresponding authors: Fan Yang; Zhaori Bi; Xuan Zeng.)*

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Digital Object Identifier 10.1109/TCAD.2025.3573228

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Inefficiency: The design space of analog circuits is vast and heterogeneous. For instance, three-stage operational amplifiers (opamps) can have ten adjustable compensation paths, with each path accommodating up to 25 different configurations, resulting in a total of 25^{10} possible behavior-level topologies [15], [16], [17]. Each topology is further associated with its own unique parametric space. This immense design space comprises both discrete and continuous subspaces, where valid solutions are sparsely and irregularly distributed. Without domain knowledge to effectively narrow down the design space, black-box approaches often suffer from excessive trial and error, making them prone to getting stuck in suboptimal solutions within a constrained computational budget.

Noninterpretability: Interpretability is a critical requirement in analog circuit design, as the designs that are interpretable to designers adhere to physical design principles, thereby reducing the risk of performance degradation and functional failure after fabrication. However, black-box approaches lack transparency in explaining the design rationale. Consequently, even when circuits identified through automated methods demonstrate superior performance, human experts may be reluctant to adopt them due to the absence of interpretability.

Recent advancements in large language models (LLMs) have created new opportunities for electronic design automation (EDA). LLMs have already been successfully applied in digital circuit design, particularly in tasks involving register-transfer-level (RTL) code, leveraging open-source Verilog datasets for training [21], [22], [23].

The potential of LLMs in analog circuit design is promising. After acquiring human expertise in analog circuit design from relevant documents, LLMs can effectively prune the design space, thereby minimizing unnecessary explorations and improving design efficiency. Additionally, LLMs can harness their inherent reasoning and planning capabilities to methodically tackle the complexities of analog circuit design. This systematic approach not only enhances the design process but also provides clear rationales for design decisions, making the process more interpretable.

Recent studies have explored the application of LLMs in analog circuit design, addressing tasks, such as circuit sizing [24], [25], netlist generation for specified topologies [26], and topology synthesis [27], [28]. LaMAGIC [27] is an LLM-based topology generation model that employs supervised fine-tuning for automated power converter design. However, since their model lacks integrated design knowledge, it requires over 100 000 circuit simulations for training, and the resulting designs lack interpretability. Artisan [28] fine-tunes the Llama2-7b model using a domain-specific dataset of 190 million tokens to create a specialized LLM for opamp design, achieving state-of-the-art performance in opamp topology synthesis.

Despite these efforts, the application of LLMs for automating analog circuit design has seen limited success, largely due to the scarcity of available data. Advanced LLMs, equipped with sophisticated reasoning and planning abilities, generally have billions of parameters and require extensive datasets—often on the order of trillions of tokens—for training. For instance, the smallest model in the Llama3 series has eight billion parameters and necessitates 15 trillion tokens for its training [29]. Furthermore, fine-tuning existing LLMs for analog circuit design necessitates substantial datasets [27], [28]. In contrast, the analog circuit design community traditionally relies on proprietary, closed-source methods. Publicly available, high-quality analog circuit designs are scarce and prohibitively expensive to collect, presenting a significant barrier to the development of domain-specific LLMs for analog circuit design.

This work aims to propose a viable method for applying general-purpose LLMs to tackle data-scarce yet knowledge-intensive tasks in analog circuit design, thereby reducing the need for training or fine-tuning domain-specific LLMs that require large amounts of specialized data. To achieve this, we propose *Atelier*, an LLM-based multiagent analog circuit design framework. Atelier automates the analog circuit design process by leveraging multiple general-purpose LLM agents in conjunction with a compact, high-quality domain-specific knowledge base. We evaluate and demonstrate the capabilities of Atelier through its application to opamps and comparators, which are among the most widely used and highly customized types of analog circuits.

The key contributions are listed as follows.

1) We propose an LLM-powered knowledge extraction approach and construct a compact, high-quality knowledge base for analog circuit design. To address data scarcity, our raw corpora include review papers and research papers on classic circuit structures. We develop design-oriented knowledge templates to focus on specific design tasks, and leverage the advanced text comprehension and summarization capabilities of LLMs to populate these templates with distilled, essential knowledge for opamp design. Each entry in the knowledge base is then rigorously reviewed by human experts to ensure accuracy. During the design process, this knowledge base supports the retrieval-augmented generation (RAG) technique, enabling Atelier to efficiently retrieve relevant information to aid in the design process.
2) We design a general-purpose LLM-based multiagent framework to facilitate automated analog circuit design. The framework consists of five LLM agents, each dedicated to a specific design step in the analog circuit design process—topology selection, circuit analysis, parameter tuning, topology modification, and design decision. A detailed chain of thoughts (CoT) is customized for each agent to break down tasks into manageable operations, thereby enhancing the instruction-following and reasoning capabilities of the LLMs. The agents are organized within a graph-of-thoughts (GoT) architecture that enables self-reflection, backtracking, and tool integration, forming a collaborative multiagent system. This system, coupled with access to the knowledge base and an optimizer, automates the analog circuit design process, ensuring interpretability through clear design rationales and enhancing both design quality and efficiency.
3) Experimental results demonstrate that Atelier significantly outperforms black-box optimization

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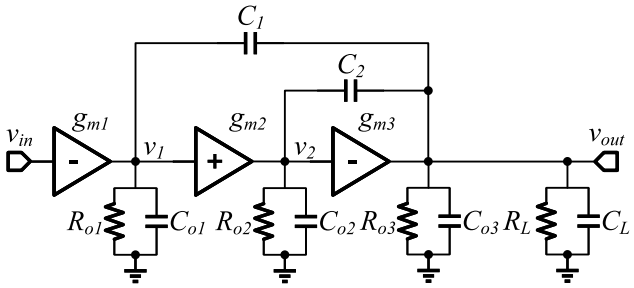


Fig. 1. Behavioral model for the NMC opamp. The main amplifier stages are composed of g_{m1}, g_{m2}, and g_{m3}, along with their associated resistors and capacitors. C_1 and C_2 are the two compensation paths. R_L and C_L denote the load resistor and capacitor, respectively.

methods [17], [19], general-purpose LLMs, such as GPT-4 [30] and GLM-4 [31], as well as the state-of-the-art LLM-based method Artisan [28], in terms of success rates, design quality, and runtime across a diverse set of design specifications.

The remainder of this article is organized as follows. Section II introduces the background. Section III demonstrates the proposed approach. Section IV presents the experimental results. Section V concludes this work.

II. Background

A. Problem Formulation

Analog circuit design can be mathematically formulated as an optimization problem, which is realized by adjusting the topology $t \in \mathcal{T}$ and its design parameters $\mathbf{x} \in \mathcal{X}$ to meet the design specifications, as shown in

$$\begin{aligned} &\max_{t\in\mathcal{T},\ \mathbf{x}\in\mathcal{X}} f(t,\mathbf{x}) \\ &\text{s.t. } c_i(t,\mathbf{x}) > c_{\text{th}}^i \quad \forall i \in \{1,\ldots,N_c\} \end{aligned} \tag{1}$$

where $c_i(\cdot)$ and c_{th}^i are the ith performance metric and its corresponding specification, respectively. N_c denotes the number of performance constraints. The objective function $f(\cdot)$ is often the figure of merit (FoM) of an analog circuit.

B. Analog Circuit Behavioral Models

We adopt the behavioral models in opamp design. In these behavioral models, the ith amplifier stage is represented by an ideal voltage-controlled current source (VCCS) g_{mi}, along with a lumped resistor R_{oi} and a lumped parasitic capacitor C_{oi} connected in parallel. As these model effectively capture the primary response of analog circuits, they are widely used in both manual and automated analog circuit design processes [15], [16], [17], [19]. Compared with transistor-level approaches, behavior-level topology synthesis not only significantly reduces the design space by abstracting away the granular details of transistor-level implementation but also reduces computational costs by avoiding complex transistor models. This enables efficient exploration of a broader range of design possibilities.

The behavioral model for a nested Miller compensation (NMC) opamp is illustrated in Fig. 1. An opamp typically consists of multiple main amplifier stages for voltage amplification and several compensation paths to ensure stability. Both the topology of the compensation paths and the parameters of all the circuit components must be carefully designed.

Transistor-level implementations of the behavioral models can be generated by using the g_m/i_d-based mapping method as described in [32]. In this process, the transistor-level circuit structure is derived from the behavior-level topology. Specifically, the amplifier stage connected to the input node is converted into a differential amplifier and the remaining stages are converted into common source amplifiers, as shown in Fig. 6(c) and (e). The transistor dimensions are then determined based on the behavior-level parameters using the g_m/i_d lookup tables.

C. Large Language Model Techniques

LLMs, built on the Transformer architecture [33], are distinguished by their substantial parameter size, enabling sophisticated natural language processing and generation. Recent advancements, exemplified by models, such as GPT-4 [30], the Llama series [29], [34], [35], and GLM-4 [31], have shown exceptional proficiency across a wide range of tasks. Additionally, several advanced techniques are frequently employed to enhance the capabilities of LLMs for domain-specific applications.

1) Prompt Engineering: Prompt engineering [36] involves carefully crafting prompt phrases to guide LLMs in accomplishing specific tasks. These adapted LLMs are often referred to as *LLM agents*. By employing prompt engineering, LLMs can perform context-relevant tasks without the need for additional training. This approach effectively leverages the model's intrinsic knowledge to achieve desired outcomes.

2) Chain of Thoughts: CoT [37] is a prompting technique that encourages the model to generate a step-by-step reasoning process, similar to how a person might break down a problem into smaller, more manageable parts. Instead of jumping directly to the answer, the model is prompted to think through the problem, articulating each step along the way. This method is particularly useful for tasks that require logical reasoning, arithmetic, or multistep problem-solving.

3) Graph of Thoughts: GoT [38] is a more advanced technique that organizes the reasoning process in a graph structure. In this approach, nodes represent different ideas, solutions, or hypotheses, and edges represent the relationships or transitions between them. GoT allows the model to explore complex, interconnected possibilities, enabling a more sophisticated and holistic understanding of the problem. We borrow concepts from the GoT approach to organize the execution and interaction of multiple LLM agents in analog circuit design, thereby facilitating and enhancing the design process.

4) Retrieval-Augmented Generation: RAG [39] is an advanced technique designed to enhance the performance of LLMs in domain-specific tasks. RAG operates by combining the generative capabilities of LLMs with the retrieval of relevant information from an external knowledge base. Specifically, it directs the LLM to query a domain-specific database, often using vector-based retrieval methods, to obtain

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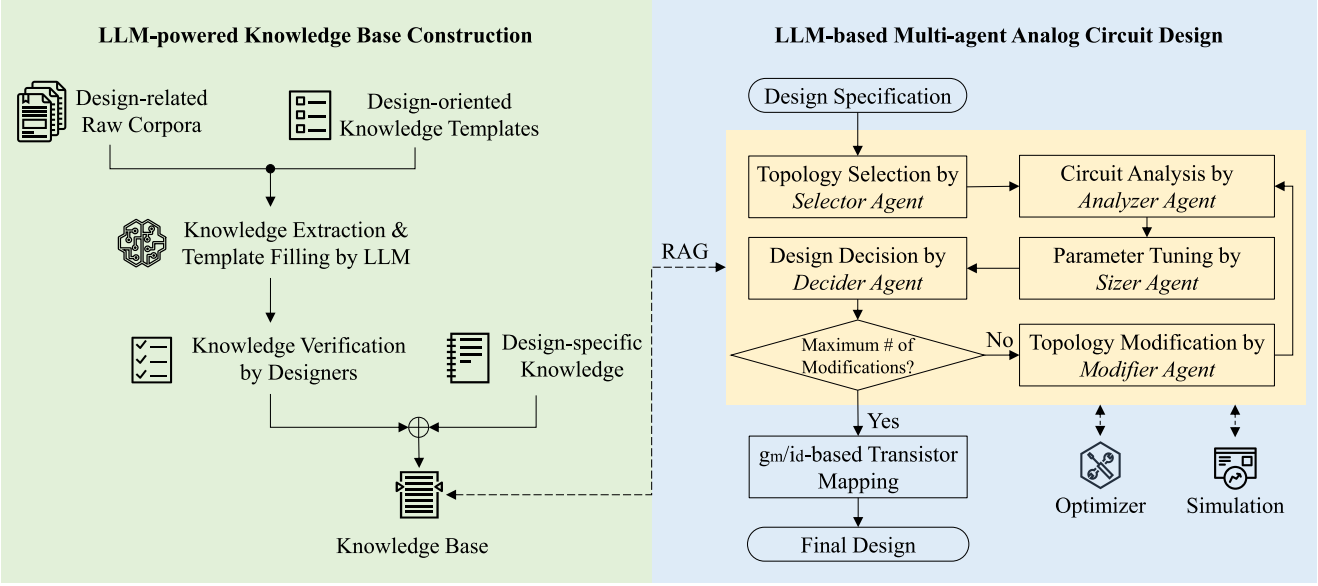


Fig. 2. Overall framework of Atelier.

pertinent content that can be integrated into the model's responses. This augmentation enables the LLM to produce more accurate and contextually relevant outputs, thereby significantly improving its effectiveness in tasks requiring specialized knowledge, such as analog circuit design.

III. Proposed Approach

This section outlines the architecture of Atelier, an LLM-based multiagent framework developed for automated analog circuit design. The framework is divided into two primary stages: 1) LLM-powered knowledge base construction and 2) LLM-based multiagent analog circuit design, as depicted in Fig. 2. In the LLM-powered knowledge base construction phase, general-purpose LLMs extract and organize relevant information from raw documents into structured templates specifically designed for analog circuit design tasks. This approach builds a compact, high-quality knowledge base that enhances the LLM's understanding of circuit design tasks through RAG. In the LLM-based multiagent analog circuit design stage, multiple LLM agents collaborate within a specialized GoT structure, enabling mechanisms like self-reflection, backtracking, and integration with an optimizer. Atelier automates the circuit design process by generating netlists from design specifications without human intervention. Additionally, it offers rationales for design decisions at each step, leveraging the reasoning capabilities of LLMs. The following sections provide detailed discussions of each stage and the design rationale behind them.

A. LLM-Powered Knowledge Base Construction

1) Motivation of LLM-Powered Knowledge Base Construction: When infusing LLMs with domain-specific knowledge, prior works often use raw corpora, such as textbooks and research papers, directly as the source for RAG [24] or as the dataset for finetuning LLMs [28]. However, this approach has limitations because the raw corpora frequently contain substantial information that is irrelevant to the specific task, leading to the retrieval of extraneous data. This can reduce the accuracy of LLM responses and the efficiency of RAG.

To address these limitations, we first distill the necessary knowledge from design-related raw corpora and then utilize RAG to integrate this knowledge into the design process. This process ensures that only the most relevant and task-specific information is retained, leading to more focused and precise retrievals and enhancing the accuracy of the LLM's responses. By refining the knowledge base, we also reduce the volume of data the RAG system must process, thus improving retrieval speed and overall efficiency. To mitigate the additional cost of knowledge extraction, we leverage the LLMs' advanced text comprehension and summarization capabilities. Furthermore, the extracted knowledge is formatted according to user-defined templates, ensuring that the content aligns with the specific design requirements. Finally, the knowledge base is constructed from verified extracted information and supplemented with a small amount of design-specific knowledge, such as parameter bounds for components in the specific design, thereby enhancing the effectiveness of analog circuit design using RAG.

2) LLM-Powered Knowledge Base Construction Flow: Our LLM-powered knowledge extraction flow is illustrated in Fig. 2. This flow begins with two inputs: 1) the design-related raw corpora, which contain the essential knowledge for analog circuit design and 2) design-oriented knowledge templates that define the structure and form of the knowledge to be extracted for use in the design process. In the next step, we utilize general-purpose LLMs to populate these templates with knowledge extracted from the raw corpora. This process leverages the LLMs' exceptional capabilities in text comprehension and summarization. The third step involves experienced designers verifying each entry in the knowledge

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TABLE I
DETAILED INFORMATION OF THE RAW CORPORA

Name	Number	Tokens (Words)
Opamp review papers	2	22857
Opamp research papers	8	29347
Circuit netlist manual	1	7768
Behavioral model descriptions	1	152
Total	**12**	**60124**

base generated by the LLMs. This verification is crucial, as LLMs can sometimes produce incorrect information due to hallucination, and inaccuracies in the knowledge could lead to design failures. Finally, the verified knowledge, supplemented with a small amount of additional design-specific information, forms a comprehensive knowledge base that serves as the source for RAG during the circuit design process. This structured and validated approach ensures that the knowledge base is both accurate and highly relevant, thereby enhancing the effectiveness of the design process.

3) Implementation Details: We provide a detailed description of the knowledge base construction process using opamps as an example. The raw corpora for opamp design are composed of three key components: 1) circuit papers; 2) circuit netlist manuals; and 3) behavioral model descriptions. These components collectively encompass the essential knowledge required for opamp design. Specifically, the opamp papers include two review papers [40], [41] that compare and summarize the advantages and disadvantages of various opamp structures, along with research papers that provide detailed descriptions of the eight most widely used structures. The behavioral model descriptions are taken from Section II-B. The detailed information of the raw corpora is listed in Table I. The total number of tokens in our raw corpora is 60 124, which is approximately 3160 times smaller that of Artisan [28] (190 million tokens), significantly reducing the effort required to collect the corpora. Furthermore, our corpora consist solely of published papers, excluding forum posts and tutorial documents that might contain inaccuracies as Artisan [28], ensuring the quality of the information included.

Three types of knowledge are required to be summarized by LLMs to aid in opamp design: 1) knowledge for topology selection; 2) knowledge for topology modification; and 3) knowledge for netlist generation. The prompts used for LLM-based knowledge extraction are illustrated in Fig. 3. The LLM effectively comprehends the raw corpora and produces well-summarized design knowledge within the required templates. This knowledge is a condensed version of the raw corpora specifically targeted toward opamp design, making it a more effective aid in the design process.

The responses generated by the LLM for all three types of knowledge are meticulously reviewed by experienced analog circuit designers. With the exception of one entry related to topology modification, all responses are deemed accurate, and the incorrect entry is manually corrected. These validated responses are then organized into three distinct sections of the knowledge base. However, some design-specific knowledge, which is crucial for circuit design, might be missing from the raw corpora yet is still necessary. For example, parameter ranges for different circuit components are essential for parameter tuning, but these can vary across technology nodes and applications. This design-specific knowledge is provided by designers to complete the knowledge base.

Prompt Template for Knowledge Extraction

Summarize the [Required Knowledge] from the [Raw Corpora] with the [Note Info]. Your response should be in JSON following the [Knowledge Template].

Prompt for Topology Selection

[Required Knowledge]: Behavior-level compensation structures and qualitative design specs for various opamp topologies
[Raw Corpora]: Opamp papers, behavioral model descriptions
[Note Info]: Generate response for opamp topologies including NMC, MNMC, NGCC, NMCF, NGRNMC, IAC, DFCFC1, and DFCFC2, considering the design specifications including load capacitance value, Gain, GBW, PM, and Power.
[Knowledge Template]: Use the name of opamp topologies as keys, with their compensation structures and qualitative design specifications as values.

Prompt for Topology Modification

[Required Knowledge]: The impact of various compensation paths on performance metrics
[Raw Corpora]: Opamp papers, behavioral model descriptions
[Note Info]: Generate response for all types of compensation paths in behavior-level opamp topologies including NMC, MNMC, NGCC, NMCF, NGRNMC, IAC, DFCFC1, and DFCFC2, considering the performance metrics including load capacitance value, Gain, GBW, PM, and Power.
[Knowledge Template]: Use the name of compensation paths as keys, with their impact on performance metrics as values.

Prompt for Netlist Generation

[Required Knowledge]: The circuit netlist of various components
[Raw Corpora]: Netlist manual, behavioral model descriptions
[Note Info]: Generate response for all types of components in behavior-level opamps, including VCCS, resistor, and capacitor.
[Knowledge Template]: Use the component types as keys, with their circuit netlist as values.

Fig. 3. Prompts for LLM-based knowledge extraction. To extract knowledge for a specific purpose, start by replacing all placeholders in the "prompt template for knowledge extraction" with the actual values of that purpose. Next, use the updated prompts in conjunction with the related raw corpora to prompt the general-purpose LLM to extract the required knowledge in certain templates.

Table II describes the composition of the knowledge base, including a knowledge snippet for each class relevant to a specific purpose. Our knowledge base is lightweight, consisting of 1082 tokens, which represents a 55.6× reduction from the raw corpora. It is also compact, encompassing the essential knowledge needed for opamp design. This knowledge base provides a source for RAG during circuit design, effectively bridging the gap between the limited domain knowledge of general-purpose LLMs and the extensive knowledge requirements of circuit design.

Our LLM-powered knowledge extraction approach offers several advantages over the method of [28], which requires the time-consuming collection of large amounts of raw corpora

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TABLE II
DETAILED INFORMATION OF THE KNOWLEDGE BASE

Knowledge Type	Usage	Tokens	Knowledge Snippet
Summarized knowledge	Topology selection	464	NMC: The compensation structure of NMC consists of a capacitor connected between node v_1 and node v_{out} and a capacitor connected between node v_2 and node v_{out}. NMC offers excellent Gain but suffers from poor power efficiency. It delivers moderate performance in GBW and PM with a small load capacitance, but its performance with a large load capacitance is poor.
Summarized knowledge	Topology modification	346	A capacitor connected between node v_1 and node v_{out}: It has no effect on Gain and Power, improves PM but worsens GBW.
Summarized knowledge	Netlist generation	201	Capacitor: C1 (NodeA NodeB) capacitor c=VAL_C1, where C1 is the name of the capacitor, NodeA and NodeB are the names of two circuit nodes that the capacitor is connected between, and VAL_C1 is the capacitance value.
Design-specific knowledge	Parameter tuning	71	A capacitor can have a value ranging from 0.1pF to 5pF.

and relies on designers manually documenting design steps and formulas for various topologies, a labor-intensive process. First, our approach is more efficient and demands significantly less setup effort. This efficiency stems from both the reduced cost of corpora collection and the lower level of design expertise required. By shifting the focus from detailed design process documentation to judgments on knowledge entries, we minimize the need for extensive human intervention. Moreover, since the LLM handles knowledge summarization, no further human effort is necessary. Second, our knowledge base is of higher quality compared to the dataset in [28], which has not been verified by designers. Our raw corpora, composed exclusively of research papers, are more reliable, and the summarized knowledge is concise, directly relevant to the design task, and free of redundancy. Designer verification further ensures the accuracy of entries, enhancing both the reliability and utility of the knowledge base for circuit design.

B. LLM-Based Multiagent Analog Circuit Design

1) Motivation of LLM-Based Multiagent Analog Circuit Design: Analog circuit design is a complex task involving multiple steps, each with distinct functions and capability requirements. For instance, topology selection demands a qualitative understanding of various circuit structures at the circuit level, topology modification requires insight into the impact of subcircuits at a finer, subcircuit level, and parameter tuning necessitates numerical optimization capabilities. Additionally, these steps must be performed sequentially, with multiple design iterations and potential backtracking until the final design meets the specified requirements. The diversity of tasks, heterogeneous capability needs, and potential backtracking in multiple design iterations pose significant challenges for using a single LLM-based agent to complete the entire design process.

To address these deficiencies, deploying multiple LLM-based agents within a collaborative framework offers a promising strategy. By specializing LLMs into distinct agents, each tailored to a specific task in analog circuit design, and carefully organizing their interactions, complex tasks can be solved more efficiently. We propose an LLM-based multiagent framework for analog circuit design, where each agent is assigned a specific role, functioning as the solver for a particular design step. This specialization minimizes knowledge interference across different stages of the design process, enabling each agent to perform its task more effectively. Each agent operates under customized instructions that guide its approach to solving the assigned step, utilizing RAG to access relevant knowledge from the knowledge base, ensuring that the agent has the necessary information to make informed decisions. Furthermore, the agents are organized using a GoT framework, which manages their interactions and supports self-reflection, backtracking, and tool integration throughout multiple design iterations. This ensures seamless collaboration among the agents to achieve the desired circuit design outcomes.

2) LLM-Based Multiagent Analog Circuit Design Flow: The design flow is depicted in Fig. 2. Inspired by the manual design process of analog circuits, it consists of five sequential steps: 1) topology selection; 2) circuit analysis; 3) parameter tuning; 4) design decision-making; and 5) topology modification. These steps are executed in a specified order, with each step performed by an LLM agent. The process begins with the *Selector Agent* choosing an appropriate circuit topology based on the target specifications. This is followed by the *Analyzer Agent*, which conducts a detailed analysis of the selected topology to generate the corresponding netlist and identify critical design parameters. Next, the *Sizer Agent* fine-tunes these parameters to meet the desired specifications. Subsequently, the *Decider Agent* evaluates the current performance metrics against those from previous iterations, and initiates backtracking if necessary to revert to the design with the best performance. If the maximum number of modification attempts has not been exceeded, the *Modifier Agent* makes further adjustments to the topology to better align the circuit with the target specifications. Circuit analysis, parameter tuning, design decision, and topology modification continue iteratively until the maximum number of modifications is reached. Finally, the g_m/i_d method is employed to convert the optimally designed behavior-level circuit into its transistor-level implementation.

The automated LLM-based multiagent framework is implemented using the open-source Langchain framework, with the analog circuit design process represented as a directed graph. In this graph, each node corresponds to an agent responsible for a specific design step, and directed edges define the sequence of execution according to the design flow. A central design state, which consolidates all critical circuit

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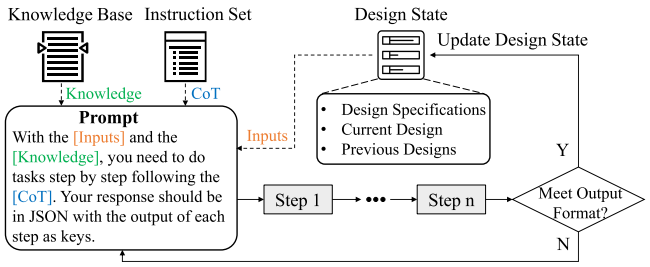


Fig. 4. Execution flow of each LLM agent in the LLM-based multiagent framework. Dashed arrows represent data flow, while solid arrows indicate the execution order. The prompt is generated by replacing placeholders in square brackets with actual values from the design state, the knowledge base, and the instruction set. If the agent's output does not conform to the required format, the execution is rerun, enabling the LLM to self-reflect.

design information—including design specifications, design structure, netlist, design parameters, parameter bounds, and performance metrics for both the current and all previous designs—is accessible to all agents. As illustrated in Fig. 4, during execution, each agent accesses the necessary inputs from the design state, retrieves relevant knowledge from the knowledge base using RAG, and follows agent-specific CoT instructions (detailed in the next section) provided by the framework. Once an agent completes its task, its output undergoes a format check. If the output conforms to the required format (i.e., it meets the JSON format and includes all the necessary output keys), it is parsed and used to update the design state. If the output does not meet these requirements, the agent's execution is rerun. This process allows the LLM to self-reflect, ensuring the correctness of its execution. The agents proceed in a predefined order dictated by the graph's edges, continuing execution until the user-defined limit on modification attempts is reached.

3) CoTs for Agents: To ensure the correct execution of each design step, which demands precise instruction following and advanced reasoning capabilities from the agents, we meticulously construct CoTs for each agent, as illustrated in Fig. 5. These CoTs guide the agents in breaking down their assigned subtasks into simpler, logically connected steps, enabling them to execute their roles effectively within the design process. The CoTs are stored in an instruction set in the form of natural language.

Selector Agent: Selecting an appropriate topology at the outset of the design process is essential, as it defines the upper limit of performance achievable through subsequent design. Upon receiving the design specifications from the user in natural language, the Selector Agent begins by qualitatively classifying each spec (e.g., large, medium, and small) using its fundamental knowledge of analog circuit design. Leveraging this qualitative assessment, the Selector Agent retrieves topology selection knowledge from the knowledge base and identifies the most suitable topology through its reasoning capabilities. Following the selection, the design structure corresponding to the chosen topology is retrieved from the knowledge base and used to update the design state.

Analyzer Agent: Analysis of the design structure provides the foundation for subsequent design steps. The Analyzer

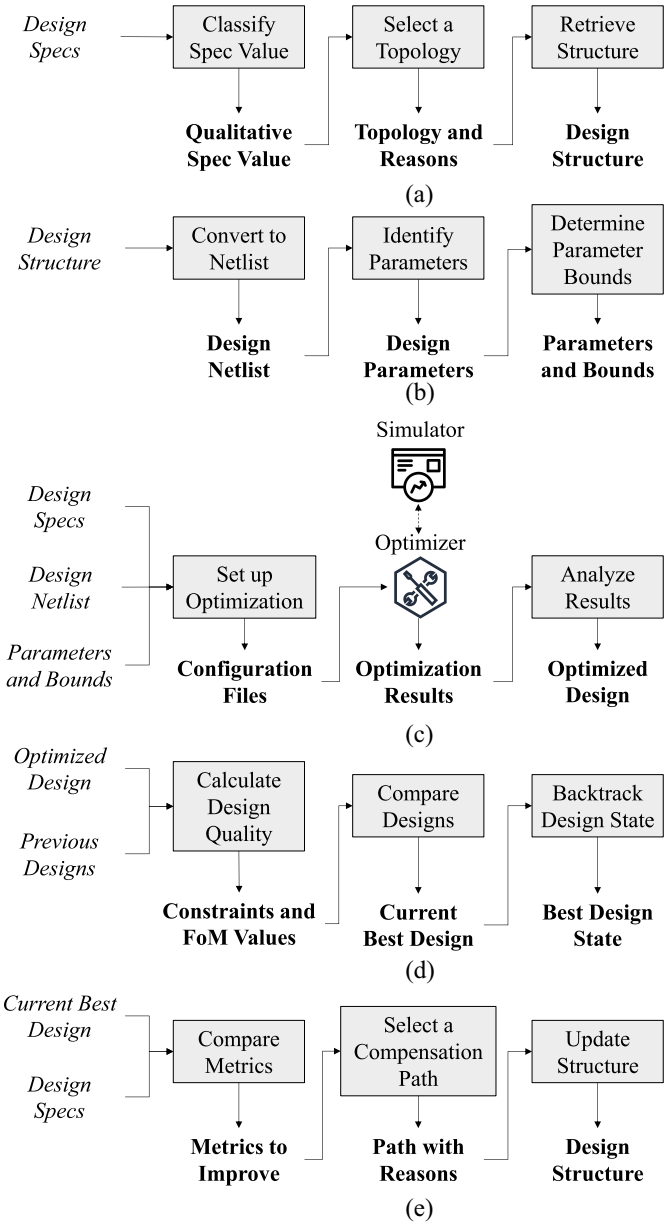


Fig. 5. CoTs for (a) topology selection by *Selector Agent*, (b) circuit analysis by *Analyzer Agent*, (c) parameter tuning by *Sizer Agent*, (d) design decision by *Decider Agent*, and (e) topology modification by *Modifier Agent*. Each design step is decomposed into simpler actions in the gray boxes. The inputs and outputs for each design step are shown in *italics* and **bold type**, respectively.

Agent begins by converting the input design structure, described in natural language, into netlist format, a format compatible with circuit simulation tools. This conversion is achieved by retrieving netlist generation knowledge from the knowledge base. Once the netlist is created, the agent identifies the design parameters that require tuning from the netlist lines. It then determines the parameter bounds using design-specific knowledge, enabling the subsequent parameter tuning.

Sizer Agent: The Sizer Agent is designed to tune circuit parameters to meet specific requirements. Given that LLMs lack the precise numerical capabilities needed for accurate parameter tuning, the Sizer Agent utilizes a numerical optimizer, specifically the covariance matrix adaptation evolution strategy (CMA-ES) [42], which is a cutting-edge

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evolutionary optimization approach based on the covariance matrix adaptation technique for constrained optimization. During the optimization process, the Sizer Agent first sets up the necessary configuration files for the optimizer, including the optimization target, constraints, parameterized netlist, design parameters, and parameter bounds. It then invokes the optimizer. The optimizer iteratively generates candidate design parameters and conducts circuit simulations to evaluate these candidates. The performance constraints are addressed using an augmented Lagrangian method [43], which progressively increases the penalty weight for constraint violations (CVs) while dynamically updating the Lagrange multipliers. This approach ensures that the optimization solution iteratively converges toward and ultimately satisfies the specified constraints. Upon completion of the optimization process, the optimized design, including the optimal performance metrics and corresponding design parameters, is retrieved from the optimization reports.

Decider Agent: Analog circuit design typically requires multiple iterations, making it essential to monitor progress, verify improvements, and revert to previous states if necessary. The Decider Agent manages this process by evaluating whether the current design improves upon previous iterations. It takes as input the design optimized by the Sizer Agent, calculates its constraints and FoM values, and compares them with those of all previous designs. If the current design does not meet the defined optimality criteria, the Decider Agent initiates backtracking, retrieving the optimal design from the history and updating the design state accordingly. While the quality of a design is currently assessed based on constraints and FoM values, our approach is highly flexible, allowing users to easily incorporate their own criteria for optimal design or manually select a design that supports an interactive design method.

Modifier Agent: If the desired specifications cannot be met through parameter tuning alone, modifying the topology becomes necessary. The Modifier Agent first compares the performance metrics of the current best design with the required design specifications. If not all the performance constraints are met, the violated performance metrics are targeted for improvement. Otherwise, efforts should focus on enhancing the circuit's FoM. The Modifier Agent then selects a compensation path that can improve the previously identified metrics by retrieving topology modification knowledge from the knowledge base, leveraging the reasoning capabilities of the LLM. The compensation paths available for selection include all paths across the eight topologies, excluding those that already exist in the current design structure and those previously attempted. Finally, the Modifier Agent incorporates the selected new path into the existing design. If the current design structure contains a path with the same connection as the new path, it is removed before the new path is added. This approach simplifies the circuit structure and eliminates potential coupling between branches, effectively enhancing the design quality.

4) Summary of LLM-Based Multiagent Analog Circuit Design: Compared to the design flow of Artisan [28], which employs a single agent and allows only one topology modification, our LLM-based multiagent framework for analog circuit design utilizes multiple LLM agents and supports multiple topology modifications, covering a broader range of circuit topologies with greater performance potential. Each agent follows its own CoT and is organized within a GoT framework, incorporating several advanced mechanisms.

Self-Reflection: Despite carefully designed prompts and customized CoTs, LLM agents sometimes produce responses that do not fully meet the design requirements, leading to failure of the design process. To enable fully automated analog circuit design, we prompt LLM agents to self-reflect on their outputs and re-execute if necessary, ensuring responses conform to the required format and specifications.

Backtracking: Since the effects of a topology modification can only be assessed after circuit sizing and simulations, backtracking becomes essential when a modification results in inferior performance compared to earlier designs. To compare the performance of different designs, we normalize the CV [44] relative to the constraint values to define the normalized CV (NCV)

$$\mathrm{NCV} = \sum_{i=1}^{N_c} \max\left(\frac{c_{\mathrm{th}}^i - c_i}{c_{\mathrm{th}}^i}, 0\right). \tag{2}$$

The design with the lowest NCV is considered optimal. If multiple designs have the same lowest NCV, the design with the highest FoM is deemed to be the optimal design. By monitoring design progress at each iteration and using the Decider Agent to backtrack to the current optimal design when needed, we ensure consistent improvement in design quality across iterations, enhancing overall performance.

Tool Integration: While LLMs excel at natural language tasks, they lack essential capabilities for analog circuit design, such as numerical optimization. To bridge this gap, we integrate external tools, specifically CMA-ES, to handle numerical optimization and invoke circuit simulations for sizing. The Sizer Agent automatically generates configuration files for CMA-ES and reads optimization results from the optimization logs, enabling seamless tool integration. Unlike Artisan [28], which relies on simplified design equations and multiple rounds of question-and-answer to determine circuit parameters, our approach reduces token usage and shortens the sizing process while delivering superior design quality.

IV. Experimental Results

This section evaluates Atelier's capability for two types of analog circuits: 1) opamps and 2) comparators.

A. Experiment Settings

Atelier is implemented in Python. The experiments are conducted on a Linux workstation with two 64-core Intel Xeon Gold 6226R @ 2.9-GHz CPUs. Circuit simulations are performed using the Spectre simulator. For Atelier, we utilize GLM-4 as the LLM for knowledge extraction and to construct multiple LLM agents for analog circuit design. The *pycma* API is employed to call the CMA-ES for parameter tuning, with the hyperparameters of CMA-ES set to their default values. We

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TABLE III
DESIGN SPECIFICATION SETS

Specs	Gain (dB)	GBW (MHz)	PM (°)	Power (μW)	Offset (μV)	Noise ($nV/\sqrt{Hz}$)	Area (μm^2)	C_L (pF)
S-1	>85	>0.7	>55	<250	<3000	<500	<250	10
S-2	>110	>0.7	>55	<250	<3000	<500	<250	10
S-3	>85	>5	>55	<250	<3000	<500	<250	10
S-4	>85	>0.7	>55	<50	<3000	<500	<250	10
S-5	>85	>0.7	>55	<250	<3000	<500	<250	1000

run the experiments for ten times to average out the random fluctuations. We evaluate the success rate (defined as the proportion of optimization runs that successfully find designs meeting all performance constraints), average optimization results of the successful runs, and average runtime.

B. Opamp Results

The SMIC 180-nm process is used for opamp design. The following metrics are considered for opamps: open-loop gain (Gain), gain-bandwidth product (GBW), phase margin (PM), power consumption (Power), input-referred voltage offset (Offset), input-referred voltage noise at 1 kHz (Noise), transistor area (Area), and a fixed load capacitance (C_L). To thoroughly assess the design capabilities of all methods, we conduct design tasks across a wide range of specifications, as detailed in Table III. S-1 defines a set of fundamental specifications for opamps, while S-2 through S-5 extend the specifications to target specialized performance metrics: S-2 prioritizes higher gain for high-precision applications, S-3 emphasizes higher GBW for medium-frequency signal processing, S-4 focuses on lower power consumption for ultralow-power sensors or IoT devices, and S-5 enhances ultralarge capacitive load driving capability for applications such as LCD drivers. The FoM [45], which holistically combines bandwidth, load-driving capability, and power efficiency, is set to be the unified optimization objective across all specifications

$$\text{FoM} = \frac{\text{GBW [MHz]} \cdot \text{C}_\text{L} \left[\text{pF}\right]}{\text{Power [mW]}}. \tag{3}$$

We set the load resistance to 1 MΩ and supply voltage to 1.8 V.

We compare our method with two state-of-the-art black-box optimization methods based on BO [17] and RL [19], two state-of-the-art general-purpose LLMs, GPT-4 [30] and GLM-4 [31], and a state-of-the-art LLM-based topology synthesis method Artisan [28]. Two variants of Atelier are evaluated for opamp design: Atelier-1 and Atelier-3, with a maximum of 1 and 3 topology modifications, respectively. The simulation budget for parameter tuning is 100 for both the initial design and each modified design. Consequently, Atelier-1 has a total simulation budget of 200, while Atelier-3 has a budget of 400.

Tables IV presents the performance results for the opamp design tasks. The results for GPT-4 and GLM-4 are omitted because they are unable to complete the design process for all specification sets. As shown in Table IV, both Atelier-1 and Atelier-3 consistently outperforms all baseline methods in terms of success rate, FoM values, and runtime. Atelier-1 exceeds the success rate of the best-performing baseline, Artisan, and Atelier-3 is the only method that achieves successful designs in all runs across all sets of specifications, including those challenging specifications where black-box methods consistently fail, underscoring its robustness. Compared with black-box methods, Atelier-1 demonstrates a 1.57× to 4.70× improvement in average FoM and a 53.8× to 87.3× runtime speedup. Compared with Artisan, Atelier-1 achieves a 1.19× to 2.63× improvement in average FoM and a 2.55× to 4.65× runtime speedup. By allowing more topology modifications, Atelier-3 generates circuits with improved FoM at the cost of a longer runtime compared to Atelier-1. Compared with black-box methods, Atelier-3 demonstrates a 2.06× to 5.33× improvement in average FoM and a 28.2× to 43.4× runtime speedup. Compared with Artisan, Atelier-3 achieves a 1.76× to 3.44× improvement in average FoM and a 1.37× to 2.74× runtime speedup.

Both black-box methods, BO and RL, exhibit instability, as evidenced by their low success rates and inefficiencies, characterized by prolonged design times. Their poor performance stems from their black-box nature, which is inadequate for exploring the vast topological space with sparse and irregularly distributed solutions. General-purpose LLMs, such as GPT-4 and GLM-4, fail to generate viable designs across all sets of specifications, highlighting their lack of domain-specific knowledge and emphasizing the need for integrating such knowledge for successful analog circuit design. Compared to Atelier-1, which allows the same number of topology modifications, Artisan demonstrates inferior performance with a significantly longer runtime. When compared to Atelier-3, which permits more topology modifications, Artisan's performance is even further diminished, while still requiring a longer runtime. Artisan's suboptimal performance can be attributed to several factors: knowledge interference caused by a single agent managing the entire design process, inaccuracies resulting from its equation-based parameter design approach, and a simplified design flow that permits only one topology modification while lacking mechanisms, such as self-reflection, backtracking, and tool integration. Furthermore, Artisan's parameter design method requires multiple rounds of question-and-answer interactions, contributing to its prolonged runtime.

Atelier's superior performance can be attributed to the incorporation of condensed, high-quality human knowledge for analog circuit design into LLMs using RAG. This enables LLMs to effectively perform topology selection, parameter tuning, and topology modification within the vast opamp design space. Additionally, the LLM-based multiagent framework, which deploys specialized agents for different design steps and organizes the agents in a GoT structure that incorporates self-reflection, backtracking, and tool integration, ensures the successful execution of the design process and significantly enhances design quality.

C. Opamp Design Examples

1) Examples of Black-Box Approaches: The black-box approaches based on BO and RL only deliver the final design without disclosing the design process to the users.

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TABLE IV
EXPERIMENTAL RESULTS ON OPAMP DESIGN

Spec	BO [17]			RL [19]			Artisan [28]			Atelier-1			Atelier-3		
	# of Suc.	Avg. FoM	Time (min)	# of Suc.	Avg. FoM	Time (min)	# of Suc.	Avg. FoM	Time (min)	# of Suc.	Avg. FoM	Time (min)	# of Suc.	Avg. FoM	Time (min)
S-1	6/10	193.9	239	3/10	127.2	200	9/10	263.9	7.84	**10/10**	597.5	**3.03**	**10/10**	**677.5**	5.94
S-2	2/10	213.1	241	1/10	187.1	193	8/10	281.3	8.34	**10/10**	335.5	**3.05**	**10/10**	**664.3**	6.00
S-3	0/10	-	242	0/10	-	177	7/10	398.0	15.31	**10/10**	680.1	**3.29**	**10/10**	**830.2**	5.58
S-4	0/10	-	236	0/10	-	175	7/10	247.6	8.07	8/10	650.6	**3.17**	**10/10**	**850.9**	5.89
S-5	1/10	11722.5	296	2/10	9371.4	192	6/10	13717.6	14.43	8/10	21367.6	**3.39**	**10/10**	**24133.5**	6.82

TABLE V
PERFORMANCE OF THE BEST OPAMPS FOR ALL SPECIFICATION SETS AFTER TRANSISTOR MAPPING

Spec	Method	Gain(dB)	GBW (MHz)	PM (°)	Power (μW)	Offset (μV)	Noise ($nV/\sqrt{Hz}$)	Area (μm^2)	FoM
S-1	Artisan	110.8	0.867	55.9	22.9	164.76	200.8	83.0	378.2
	Atelier-3	114.3	19.400	74.6	150.6	23.48	132.7	62.4	**1287.9**
S-2	Artisan	115.0	1.751	65.3	48.6	543.69	218.7	181.1	360.0
	Atelier-3	116.5	13.164	71.5	131.6	60.31	135.7	70.8	**999.8**
S-3	Artisan	106.6	5.859	89.0	108.7	2446.06	447.0	202.3	539.0
	Atelier-3	110.7	20.540	68.3	152.1	21.64	130.7	62.3	**1350.4**
S-4	Artisan	104.3	0.717	65.3	27.2	253.07	211.7	51.0	263.1
	Atelier-3	102.3	7.569	66.5	36.5	95.11	146.3	49.8	**2075.9**
S-5	Artisan	86.3	3.458	57.6	112.7	2401.22	492.1	56.4	30683.2
	Atelier-3	99.1	3.821	63.9	114.2	28.62	134.3	57.0	**33458.8**

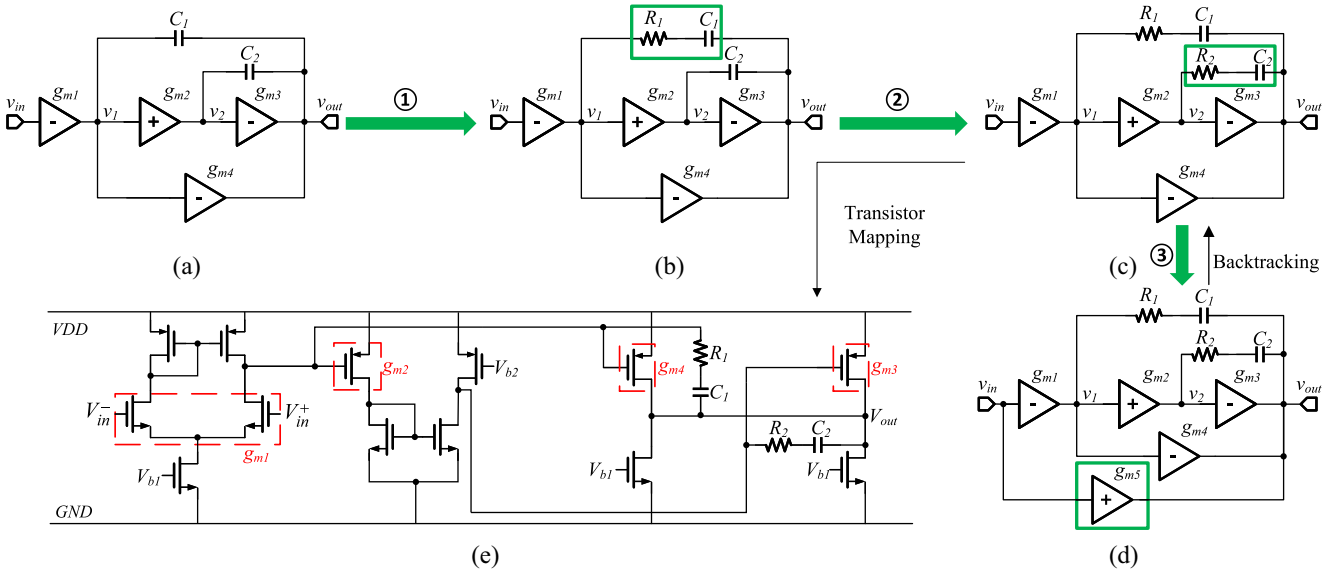


Fig. 6. Design trajectory of the best opamp for specification S-4 from Atelier-3. (a)–(e) shows the initial topology, the topology after the first modification, the topology after the second modification (also the optimized behavior-level topology), the topology after the third modification, and the final design, respectively. Each green arrow represents a topology modification, pointing from the original topology to the modified topology. The number associated with each green arrow indicates the modification number. The modified paths are surrounded by the green boxes. For brevity, the resistors and capacitors associated with the amplifier stages, the bias circuits in the final design, as well as the load resistor and capacitor are omitted.

These methods emphasize performance, often overlooking the complexity and interpretability of the resulting topologies. Consequently, the designs they produce frequently include functional units that are difficult to interpret (e.g., series-connected amplifier stages and resistor/capacitor). This lack of transparency leads to hesitancy among designers in adopting these methods.

2) Examples of General-Purpose LLMs: Given the absence of a dedicated GoT for analog circuit design in general-purpose LLMs, we manually decompose the design task into the same steps as Atelier and provide the inputs for each step. We evaluate GPT-4 and GLM-4, using the same design prompts as Atelier, with the exception of design knowledge, which is unavailable in these models due to the lack of a domain-specific knowledge base. Both LLMs exhibit similar issues, as detailed below.

For topology selection, both GPT-4 and GLM-4 consistently suggest using the NMC topology across all tested specifications, likely due to their limited knowledge of other opamp topologies. Moreover, they produce incorrect descriptions of the NMC structure. Specifically, GPT-4 incorrectly places the compensation capacitor between node v_1 and v_2 instead of between node v_1 and v_{out}, while GLM-4 omits the compensation capacitor connected between node v_2 and node v_{out}. In circuit analysis, both LLMs generate netlists containing syntax errors, and the identified design parameters are either too broad, encompassing parameters that cannot be tuned, or too limited, missing critical parameters that should

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be tuned. For parameter tuning, both models rely on an inefficient trial-and-error approach due to their lack of numerical optimization capabilities. When modifying topologies, the recommendations from both LLMs are often too vague to be implemented practically, such as suggesting "adjustment of the feedback network," or are fundamentally incorrect. For instance, GPT-4 mistakenly recommends adding "Miller compensation" to the NMC topology that already contains Miller compensation.

These shortcomings underscore the importance of a domain-specific knowledge base to adapt general-purpose LLMs for complex, domain-specific tasks like analog circuit design.

3) Examples of Artisan: Artisan employs a simplified design flow that permits only one topology modification to the initial design and does not allow backtracking. As a result, it tends to generate designs that closely resemble the initial designs and struggles to handle more complex design processes that require multiple iterations to achieve high-performance outcomes.

Furthermore, Artisan occasionally fails to complete the design process due to incorrect responses. Common errors include generating incorrect netlists during circuit analysis (e.g., omitting compensation paths), using incorrect design formulas during parameter tuning (e.g., confusing formulas between topologies or using mathematically incorrect formulas), and providing overly vague modification suggestions that are not directly implementable (similar to issues observed with general-purpose LLMs). These errors contribute to Artisan's lower success rate and limit its practical utility.

4) Examples of Atelier: We present a concrete design example from Atelier-3 to illustrate the design process. The design trajectory for the best opamp targeting S-4 is shown in Fig. 6. Initially, the NMCF architecture is selected [see Fig. 6(a)] due to its excellent low-power operation as well as good GBW and PM at a small load capacitance. However, after parameter tuning, the NMCF fails to meet the GBW requirements. To address this, the topology is modified by replacing the capacitor between node v_1 and node v_{out} with a series-connected resistor and capacitor [see Fig. 6(b)], followed by parameter tuning. The modified design meets all performance requirements. In the next attempt to further improve the FoM, the capacitor between node v_2 and node v_{out} is replaced with a series-connected resistor and capacitor [see Fig. 6(c)]. As the series-connected resistor and capacitor enhances GBW through pole-zero cancellation, this modification leads to an FoM improvement after another round of parameter tuning. The Modifier Agent then suggests adding an amplifier stage between node v_{in} and node v_{out} [see Fig. 6(d)]. This modification worsens the NCV, necessitating a backtrack to the previous design iteration. Finally, the optimized design is mapped to the transistor level following the transistor mapping method described in Section II-B, and the final design [see Fig. 6(e)] successfully meets the S-4 requirements.

Responses from Atelier at different design steps during the design for S-4 are illustrated in Fig. 7. At each design step, a specialized agent generates coherent responses to the prompt in the required format, guided by its customized CoT. These responses are generated by leveraging essential input from the current design state and retrieving relevant knowledge from the knowledge base using RAG, demonstrating Atelier's strong instruction-following and reasoning capabilities. The entire design process is carried out by multiple agents within a GoT framework that includes multiple design iterations, as well as essential mechanisms, such as self-reflection, backtracking, and tool integration. These elements collectively ensure design interpretability and enhance the overall quality of the design.

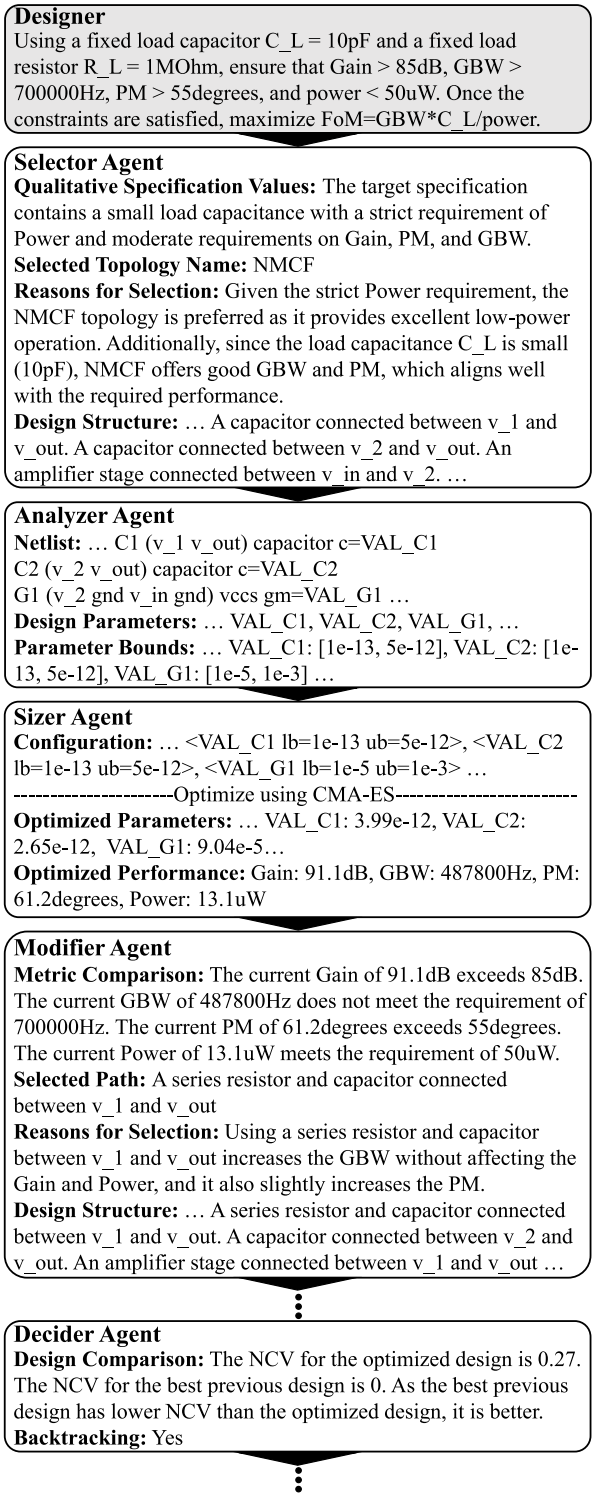


Fig. 7. Responses from Atelier to the designer's requirements during the design process for the best opamp for S-4.

The best behavior-level topologies for the remaining specifications from Atelier-3 are depicted in Fig. 8. The

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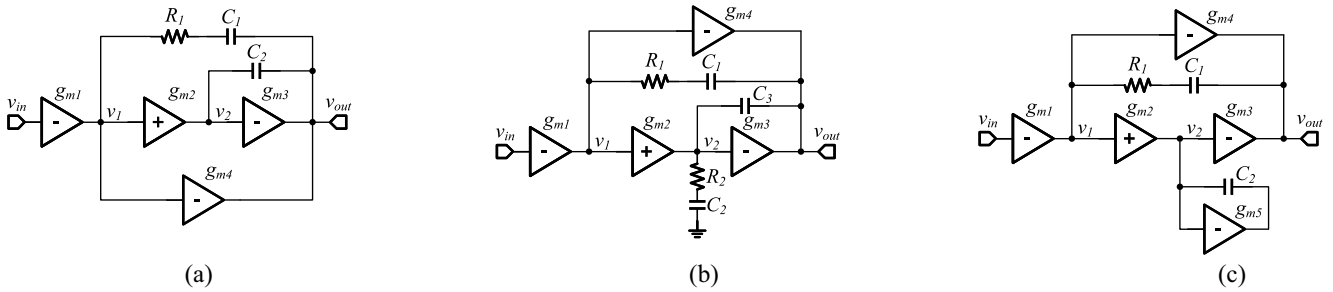


Fig. 8. Best opamp topology for (a) S-1 and S-2, (b) S-3, and (c) S-5 from Atelier-3 at the behavior-level. For brevity, the resistors and capacitors associated with the amplifier stages, as well as the load resistor and load capacitor, are omitted.

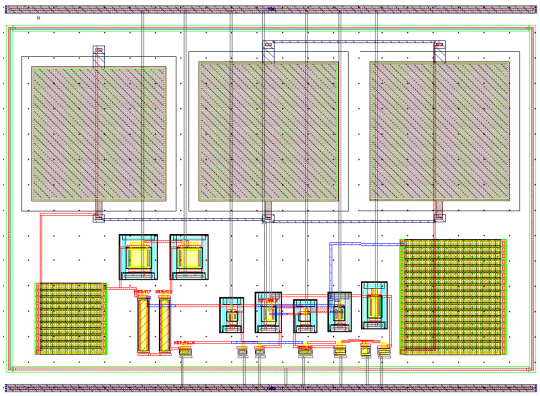


Fig. 9. Layout of the best opamp for S-4 from Atelier-3.

TABLE VI
POST-LAYOUT PERFORMANCE OF THE BEST OPAMPS FROM ATELIER-3

Spec	Gain (dB)	GBW (MHz)	PM (°)	Power (μW)	Offset (μV)	Noise ($nV/\sqrt{Hz}$)	FoM
S-1	114.2	18.557	74.3	150.5	17.07	132.7	1232.8
S-2	116.1	12.617	71.4	131.6	66.97	135.6	958.7
S-3	110.6	19.568	67.6	152.0	65.47	130.6	1287.4
S-4	103.4	7.513	64.2	39.0	58.23	146.3	1928.9
S-5	99.4	3.662	62.2	114.3	58.27	134.3	32038.5

performance of the best opamps identified by Artisan and Atelier-3 for all specification sets, after transistor mapping, are summarized in Table V. Atelier consistently generates circuits with a higher FoM, achieving improvements of 1.09× to 7.89× compared with Artisan across all specification sets.

To further validate the opamps generated by Atelier, we employ the method of [17] to automatically generate the layouts for the best opamps produced by Atelier-3 and conduct post-layout simulations. Due to space constraints, only the layout of the best opamp for S-4 is presented in Fig. 9. The post-layout performance of these opamps is summarized in Table VI. Although the layouts are not manually optimized, the parasitics remain well-controlled. The post-layout simulations reveal only minor degradations in GBW and PM due to layout parasitics, resulting in a slight reduction in the FoM compared to the transistor-level performance. These results demonstrate the effectiveness of Atelier in producing practical designs.

D. Comparator Results

The comparator is designed using the SMIC 40-nm process, with the following key performance metrics evaluated: average power consumption (Power), input-referred voltage offset (Offset), and comparator delay (Delay). The comparator operates under a 1.8-V supply voltage at a clock frequency of 1 GHz. It is designed for high-speed, energy-constrained applications. The minimization of the Power-Delay product ensures an optimal balance between energy efficiency and speed. The target specifications are outlined as follows:

$$\begin{aligned} &\text{maxFoM} = -\text{Power} \times \text{Delay} \\ &\text{s.t. Power} < 100\ \mu\text{W}, \text{Delay} < 100\ \text{ps}, \text{Offset} < 10\ \text{mV}. \end{aligned} \tag{4}$$

Since optimization methods based on BO [17] and RL [19], as well as the LLM-based method Artisan [28], cannot be readily extended to comparator design, we compare our approach with GPT-4 [30] and GLM-4 [31]. Given the significant variations in comparator topologies, we omit the topology modification step in the comparator design flow. The simulation budget allocated for parameter tuning is set to 100 for the selected design. As a result, Atelier operates with a total simulation budget of 100.

TABLE VII
RESULTS ON COMPARATOR DESIGN

Methods	# of Suc	Avg. FoM ($W \cdot fs$)	Time (min)
GPT-4 [30]	0/10	-	-
GLM-4 [31]	0/10	-	-
Atelier	**10/10**	**-3.27**	**16.93**

TABLE VIII
BEST COMPARATOR FROM ATELIER

Power (μW)	Delay (ps)	Offset (mV)	FoM ($W \cdot fs$)
84.88	35.18	4.02	-2.99

Table VII summarizes the performance results for the comparator design tasks. Notably, GPT-4 and GLM-4 fail to generate functional netlists capable of passing circuit simulation due to their limited domain-specific design knowledge. In contrast, Atelier consistently achieves successful designs across all runs. Analysis of Atelier's design process reveals that the Strong Arm latch is selected as the initial topology due to its suitability for high-speed operation, structural simplicity, and ability to simultaneously minimize delay and reduce power consumption. The performance metrics of the comparator designed by Atelier, which achieves the highest FoM, are detailed in Table VIII.

V. CONCLUSION

This article presents Atelier, an LLM-based multiagent framework to tackle the unique challenges of analog circuit

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design, which are characterized by data scarcity and the necessity for specialized domain knowledge. Atelier circumvents the need for data-intensive LLM training by leveraging general-purpose LLMs in conjunction with a carefully curated, compact knowledge base. This knowledge base is constructed using the LLMs' text comprehension and summarization capabilities. Atelier utilizes multiple specialized LLM agents organized within a GoT architecture that incorporates self-reflection, backtracking, and tool integration. Each agent is responsible for a distinct design step, following a customized CoT and retrieving relevant information from the knowledge base using RAG. Experimental results validate the effectiveness and interpretability of Atelier, demonstrating higher success rates, superior circuit performance, and shorter runtimes across various design scenarios. Concrete design examples underscore its efficacy and versatility for analog circuit design.

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